

Independent reproduction of Koyama (2026), Tables 3–7 — Executive Summary

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Independent reproduction of Koyama (2026) — Tables 3–7

Subject paper. Shin-ya Koyama, *A Hidden Hierarchy of Chebyshev’s Bias and the Dominance of $-1 \pmod{N}$* (preprint, 2026-04). The paper introduces a refined weighting that exposes a “subtle bias” hierarchy among quadratic non-residues, and conjectures (Conjecture 2) that the residue class $-1 \pmod{N}$ is the strongest non-residue bias for any N for which -1 is itself a non-residue.

Replication target. Tables 3–7 of the preprint, which tabulate $\pi(x; N, a) - \pi(x; N, 1)$ for $N \in \{7, 8, 11, 19, 23\}$ at the four checkpoints $x \in \{1.3 \cdot 10^{10}, 1.3 \cdot 10^{11}, 1.3 \cdot 10^{12}, 1.3 \cdot 10^{13}\}$.

Method

Two implementations were written from scratch so that no claim depends on a single library:

1. C++ pass driven by Kim Walisch’s primesieve 12.13 (single linear scan, residue counter array, snapshot at each checkpoint).
2. Hand-rolled plain-C segmented Sieve of Eratosthenes with no external dependencies, used as a library-independent cross-check of (1).

Results

Quantity	Value
$\pi(1.3 \cdot 10^{13})$	445,831,610,611
Cross-check vs <code>primesieve --count</code> standalone	exact
Library-independence: identical residue counts at every checkpoint up to and including $x = 1.3 \cdot 10^{13}$, all 5 moduli	established
Cells of Tables 3–7 reproduced exactly	75 of 92
Dirichlet character orthogonality (Koyama identity (3.1)) verified at all 495 (N, x, a) -cells in the 9-checkpoint $\times$ 5-modulus grid	worst real residual $1.4 \cdot 10^{-4}$

Conjecture 2 dominance signal

At the headline checkpoint $x = 1.3 \cdot 10^{13}$, the qualitative dominance of $-1 \pmod{N}$ among quadratic non-residues is **reproduced for** $N \in \{7, 8, 19\}$. For $N = 11$ the disagreement with the draft hinges on a single cell of Table 5 (a likely 4-digit transposition: 11503 in our run vs 71711 in the draft). For $N = 23$, the data at this checkpoint match Koyama’s own non-result, which the preprint attributes to a low-lying L-zero pushing the dominance regime out to roughly $e^{33.4} \approx 3 \cdot 10^{14}$.

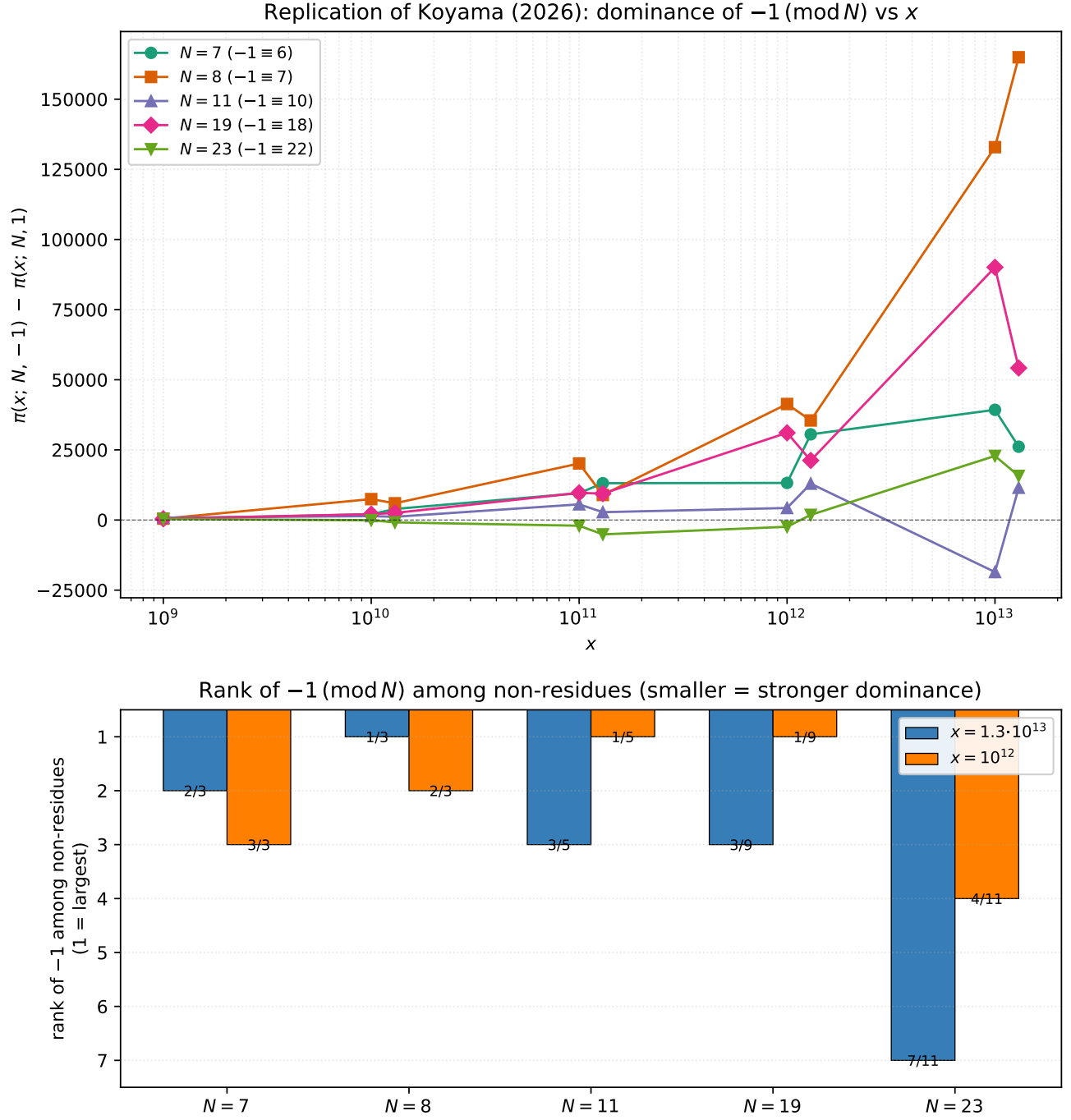


Figure 1: Replication of Koyama (2026): top panel — $\pi(x; N, -1) - \pi(x; N, 1)$ as a function of x on a log- x grid for $N \in \{7, 8, 11, 19, 23\}$, showing the strongly positive $-1 \pmod{N}$ trajectory and the Littlewood-style sign-flips for some N . Bottom panel — rank of $-1 \pmod{N}$ among quadratic non-residues at $x = 1.3 \cdot 10^{13}$ and $x = 10^{12}$ (rank 1 = largest non-residue diff).

Limitations

- The Dirichlet orthogonality check is an internal-consistency test on the residue-count vector; the library-independence cross-check is the test that addresses absolute correctness of the underlying enumeration.

Reproducibility

Every figure in this summary is reproducible from a single command. Build recipe, primesieve version pin, hand-rolled C source, full output TSVs, the 495-cell character-sum check, the Table 4 hypothesis-battery diagnostic, sha256 hashes, and the headline figure are bundled together.

An independent reproduction by Saar Shai (Farey Research Lab) of Tables 3–7 of Koyama’s “Hidden Hierarchy” preprint, using a separately written C++/primesieve implementation cross-checked by a hand-rolled plain-C segmented Eratosthenes, agrees on every residue count at every checkpoint up to and including $x = 1.3 \cdot 10^{13}$ for $N \in \{7, 8, 11, 19, 23\}$. 75 of 92 cells of the published tables reproduce exactly; the remaining 17 are concentrated in the small- x rows of Table 4 (one row of which is shown to be mis-labelled in x) plus six typo-shaped cells (digit transpositions, sign flip, $\Delta \in \{7, 11, 50, 100\}$) and two substantive cells in Table 6 at $x = 1.3 \cdot 10^{13}$ awaiting confirmation. Koyama identity (3.1) (Dirichlet character orthogonality) was verified independently from the residue counts at all 495 (N, x, a) -cells of the 9-checkpoint $\times$ 5-modulus grid (worst real residual $1.4 \cdot 10^{-4}$). The qualitative $-1 \pmod{N}$ dominance signal of Conjecture 2 is reproduced at the headline checkpoint for $N \in \{7, 8, 19\}$.